

LOT Pager Plan

LOT® Pager — Plan (v0.1)

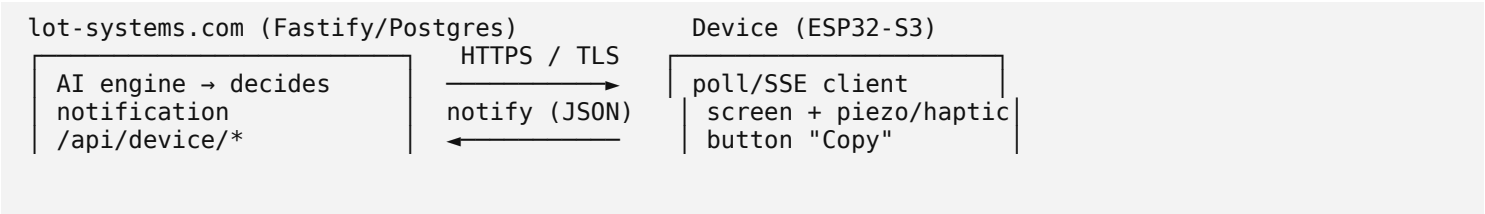
1. Product in one paragraph

A flat, square stainless-steel object. One face polished mirror steel; the other carries a small screen, a camera and one button. It sits on a Qi pad. When the LOT site (lot-systems.com) decides something is worth interrupting you for (“Coffee time!”), the device shows it and buzzes like a pager. Pressing the button acts as “Copy”: it signals back and an entry appears in the site’s **Log** tab. It also reads the room (weather-class sensors) and reports it to LOT.

2. Requirement trace (your 19 points)

#	Requirement	Where addressed
1	PCBWay	BOM §Fab, ROADMAP P3/P5 — PCB + SMT assembly + CNC at PCBWay
2	Pager-like notifications from AI site	SOFTWARE-CONNECTOR §3, FIRMWARE §4
3	2-part stainless body	§4 below
4	Flat silver 4×4 cm × 5 mm	§4 — 5 mm infeasible in v1; see §5
5	Camera	OV2640 (v1), FIRMWARE §6
6	LOT API connector	SOFTWARE-CONNECTOR
7	PDF manuals	pdf / built from MANUAL-USER.md
8	Compress info each session	README STATE block + session reports
9	Firmware documents	FIRMWARE.md
10	Software to connect with firmware	SOFTWARE-CONNECTOR §5 (host CLI)
11	Separate documents	one file per concern
12, 19	Charger / wireless charger	Qi receiver + Qi pad (BOM)
13	100-unit run	BOM §cost, ROADMAP P5
14, 15	Weather + AI-grade off-the-shelf sensors	BME688 (BSEC gas-AI), VEML7700, IMU
16	Button “Copy” → Log tab	SOFTWARE-CONNECTOR §3.3
17, 18	Polished back; camera+screen+button front	§4

3. Architecture



- Device is a thin client: it never runs AI. Server owns decisions, rate limits, and quiet hours.
- Auth: per-device token minted by a claim code the user enters on the site (SOFTWARE-CONNECTOR §2).
- Transport: HTTPS long-poll (battery-friendly, works through any NAT), upgrade to SSE later.

4. Mechanical (2-part stainless)

- Material: 316L (skin-safe, polishable) or 304 (cheaper). Recommend 316L for the polished face.
- Part A “mirror”: 40×40 mm plate, mirror polish ($R_a < 0.05 \mu\text{m}$). Qi charges through it only if thin and low-permeability — **austenitic steel attenuates Qi strongly**. Plan: put the Qi coil against a non-metal window or thin ($\leq 0.3 \text{ mm}$) section, or make the *front* part carry a plastic/ceramic charging window. This is the #1 mechanical risk (R2).
- Part B “face”: 40×40 mm tray with display window, camera aperture (sapphire/glass), button hole. Parts join with 4 hidden screws or laser weld + gasket.
- Silver look: bead-blast or satin on Part B, mirror on Part A.

5. Height analysis (the 5 mm problem)

Layer	mm
Part A plate	0.8
Ferrite + Qi coil	0.8
Thin LiPo (e.g. 3 mm cell)	3.0
PCB + components	1.2 (module is 3.1 alone)
Display module (1.28" round)	2.0
Part B front/ window	0.8
Camera module (lens stack)	≥ 3.5 (bump)
Total (v1 estimate)	$\approx 9\text{--}10 \text{ mm}$

Conclusion: v1 target $\leq 10 \text{ mm}$; 5 mm needs bare-die display, a custom flex PCB, a $\leq 1.5 \text{ mm}$ cell, and a pinhole-class camera: a v2 research item. Decision needed from S-2.

6. Risks

ID	Risk	Mitigation
R1	5 mm height	Stage: v1 $\leq 10 \text{ mm}$, v2 5 mm
R2	Qi through steel	Charging window or contact pads (pogo) fallback
R3	Wi-Fi antenna inside steel box	Use antenna aperture/plastic window; module with external antenna keepout
R4	Battery life with camera + Wi-Fi	Deep sleep, camera off by default, poll 30–60 s
R5	Privacy of camera	Hardware shutter or LED + explicit capture only
R6	Heat from charging → sensor reads high	Compensate in firmware; place BME688 away
R7	Radio certification (FCC/CE) for sale	Use pre-certified module; budget lab

ID	Risk	Mitigation
R8	Li-ion in sealed steel	UN38.3, protection IC, vent consideration